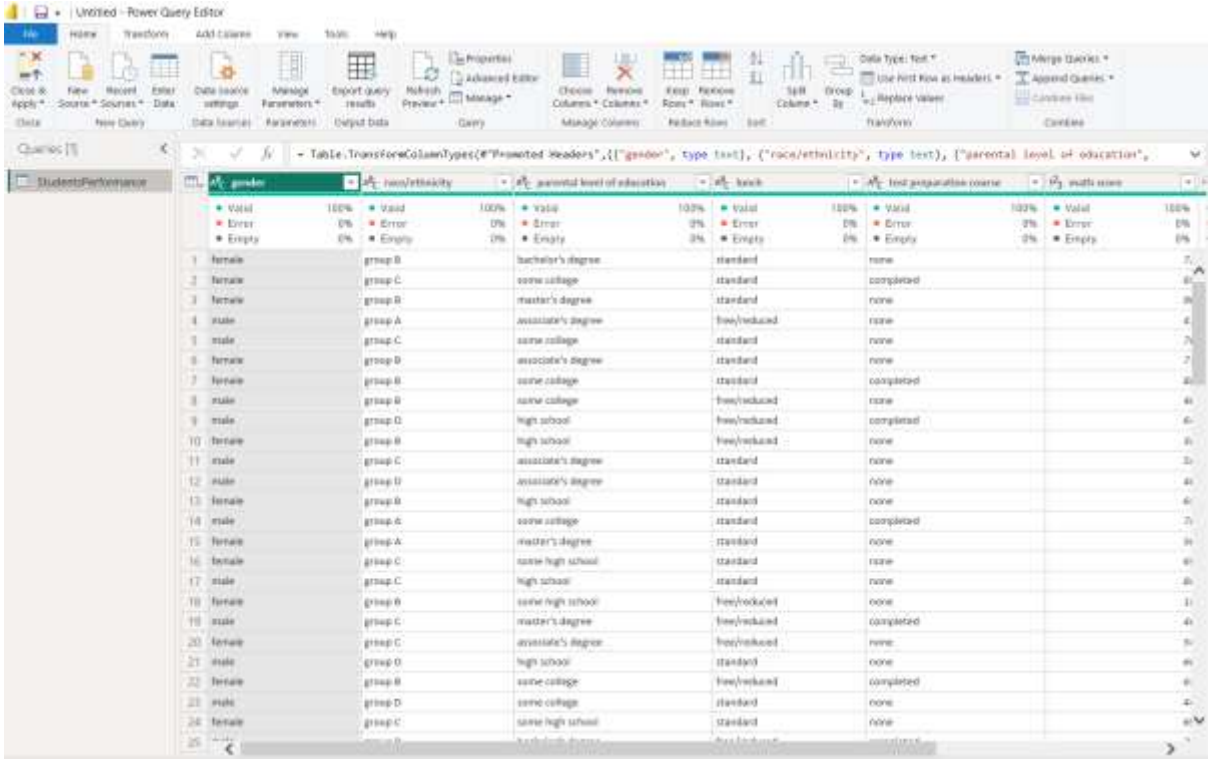
Week 7 Report — Academic Performance Data Modeling

Step 1: Loading the Original Dataset

- The dataset was first loaded into **Power Query Editor**.
- Column names were set properly, data types were corrected (text, numbers, dates), and values were checked to ensure consistency.

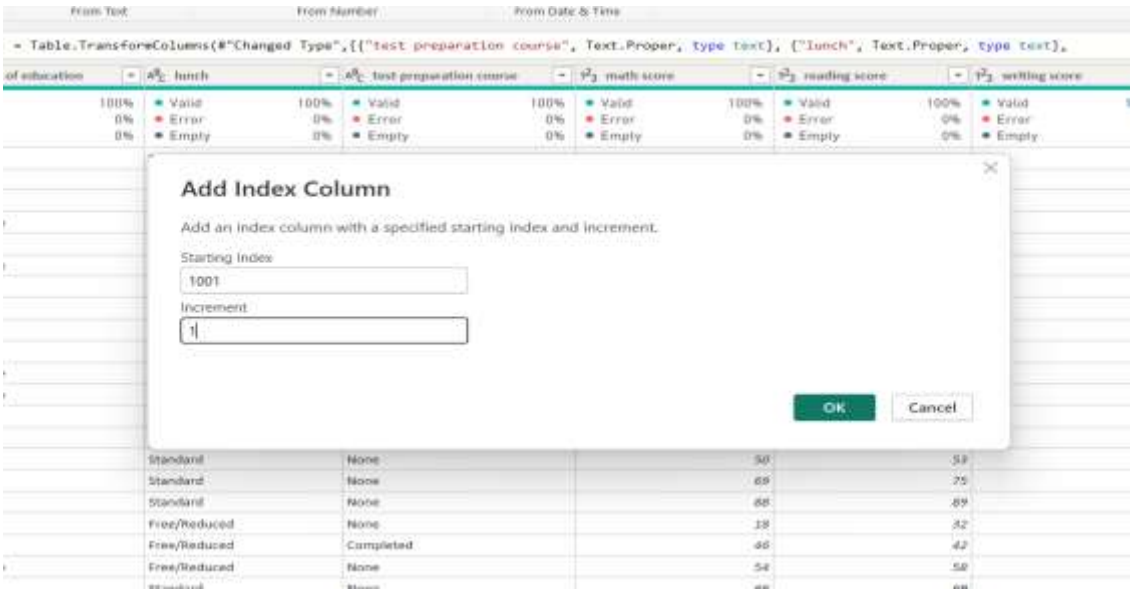
👉 This step ensured the dataset was clean and ready for transformations.



Step 2: Adding Student ID

- An Index Column was added to uniquely identify each student.
- This column was renamed as Student ID.

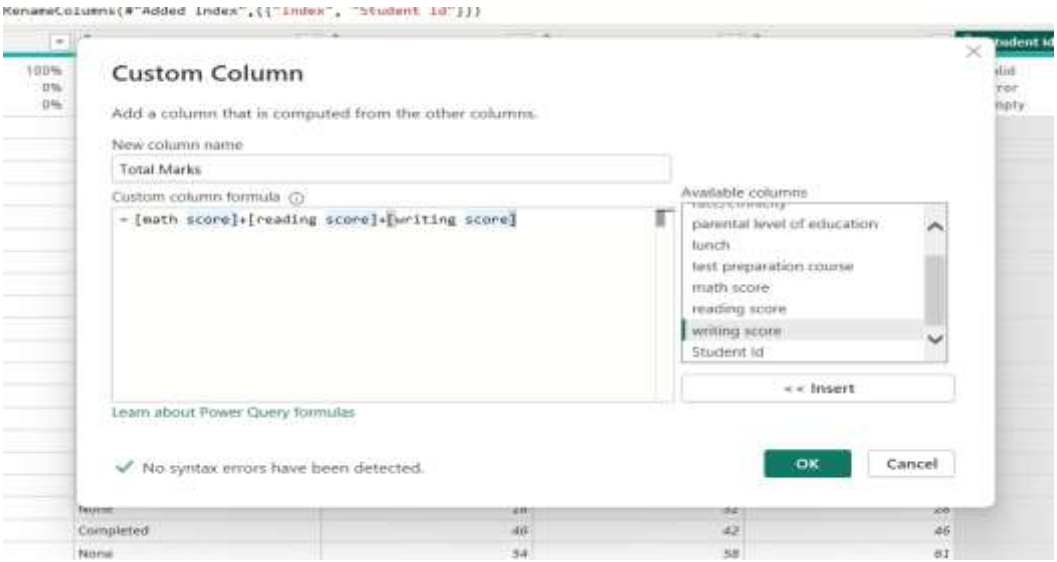
👉 Student ID became the key field for linking tables later in the model.



Step 3: Creating Total Marks

- A Custom Column was added:
  - Total Marks = Write Score + Read Score + Math Score.

👉 This gave the combined score for each student across all subjects.

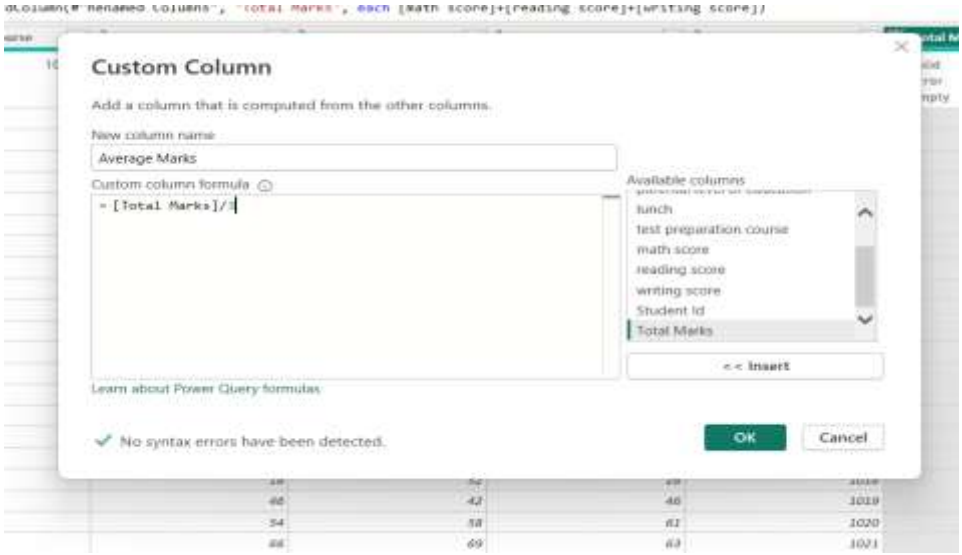


Step 4: Creating Average Marks

- Another **Custom Column** was added:

- Average Marks = Total Marks / 3.

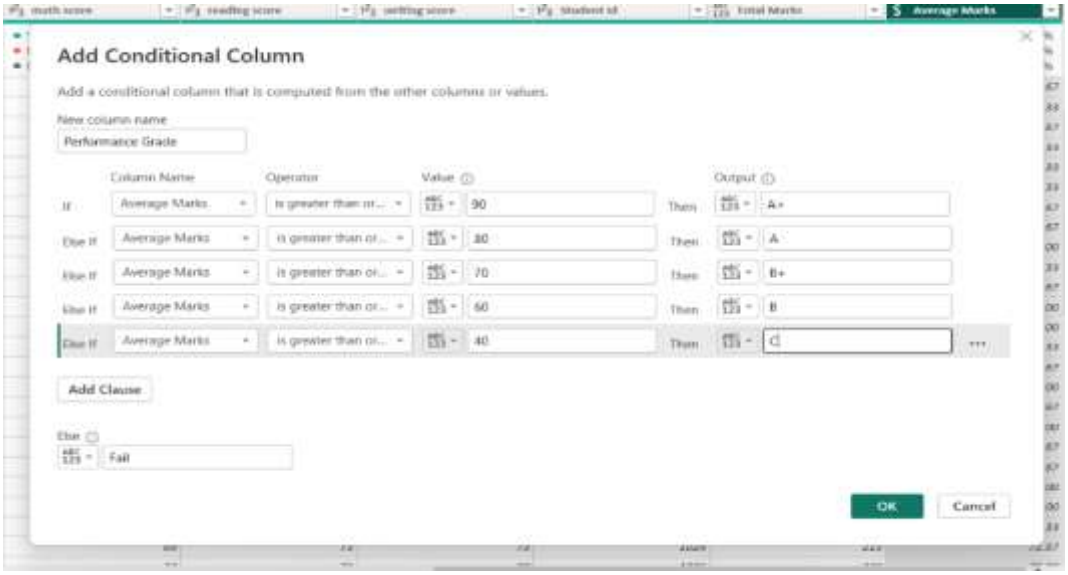
👉 This calculated the overall average performance per student.



Step 5: Adding Performance Grade

- A **Conditional Column** was created to assign grades:
  - If Average Marks ≥ 90 → A+
  - If 80–89 → A
  - If 70–79 → B+
  - If 60–69 → B
  - If 50–59 → C+
  - If 40–49 → C
  - If < 40 → F

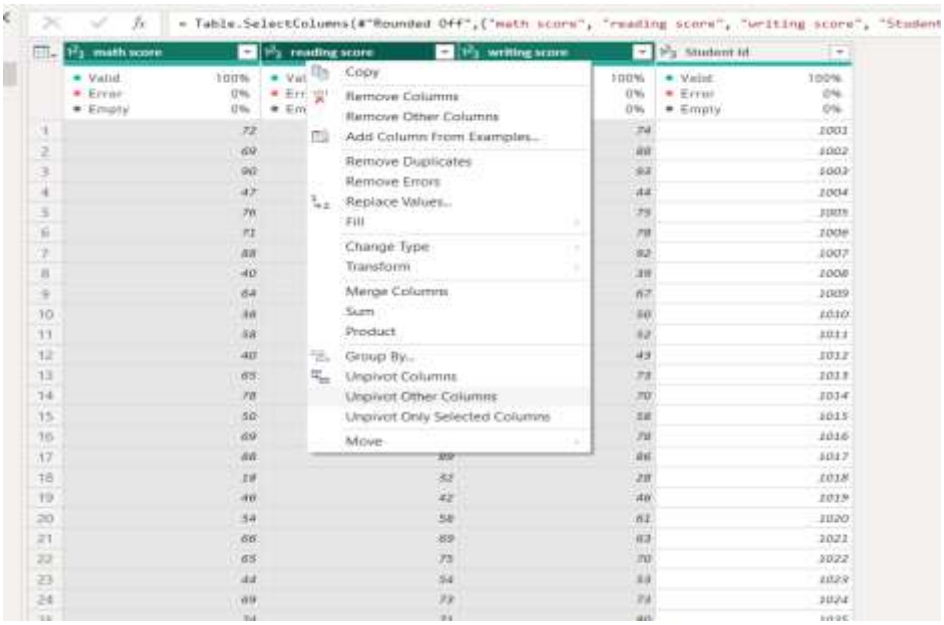
👉 This classified students into performance bands.

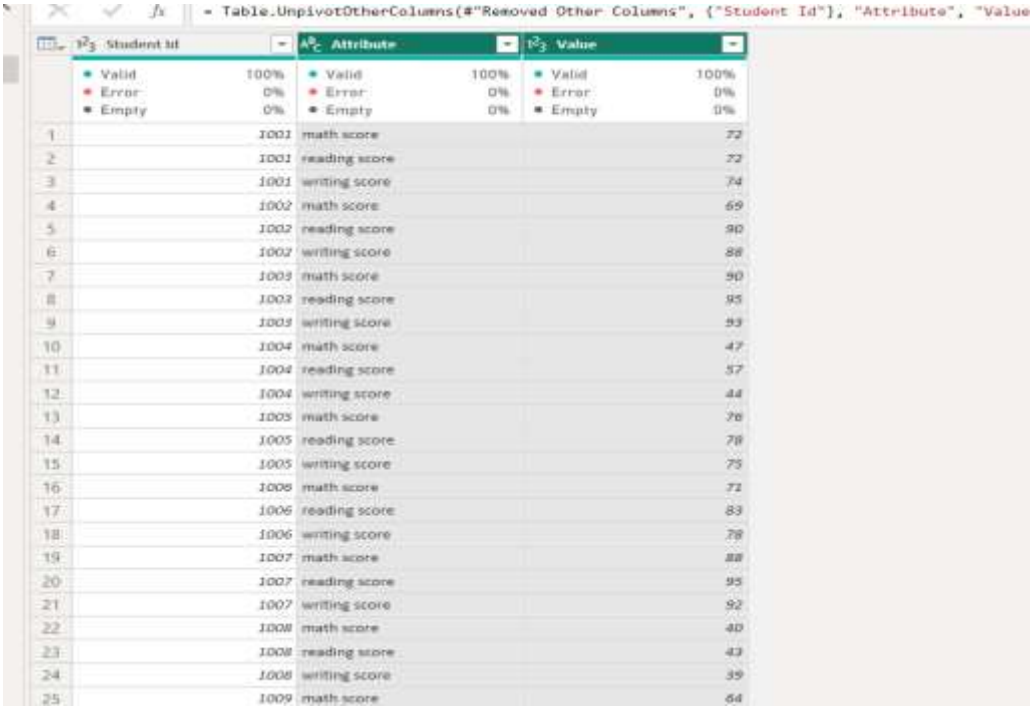


Step 6: Creating the Student Marks Table

- The original dataset was **duplicated** and renamed **Student Marks**.
- All columns except StudentID, Math Score, Reading Score, Writing Score were removed.
- Math, Reading, and Writing columns were **Unpivoted** → producing StudentID | Subject | Score.

👉 This format allows subject-wise analysis.





Step 7: Creating the Student Details Table

- Another duplicate was made, named **Student Details**.
- Only demographic columns were kept:
    - StudentID, Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation.

👉 This table was used for demographic and socioeconomic analysis.

Step 8: Creating the Performance Grade Table

- A third duplicate was made, named **Performance Grade Table**.
- Columns included:
    - StudentID, Total Marks, Average Marks, Performance Grade, Result (Pass/Fail).

👉 This table was used for grade distribution and pass/fail analysis.

Step 9: Relationships in Model View

- In **Model View**, relationships were automatically generated:
    - **Student Marks** ↔ **Student Details** ↔ **Performance Grade Table** via StudentID.

👉 This ensured all tables were connected for cross-filtering in visuals.

